

KUTLUHAN AYGÜZEL

Software Engineer — AI

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EXPERIENCE

AI Engineer, Semper Tech

Sep 2025 – Present

Istanbul, Türkiye

- Co-developed and shipped a production agentic AI system for BEKO with 6+ agent tools, running on a self-hosted GPT-OSS-120B served with vLLM on NVIDIA A100s, with Llama Guard safety filtering, Docker containers and FastAPI REST microservices.
- Rebuilt the BEKO agent's request-routing layer: the original design fine-tuned a BERT intent classifier feeding deterministic code that extracted parameters and invoked tools, but t-SNE embedding visualization and confusion-matrix analysis showed semantically overlapping intents the classifier could not separate; replaced it with LLM-driven routing and parameter extraction (native tool calling), raising tool-selection accuracy by 15+ percentage points across 6+ tools at the cost of higher token usage and negligible added latency.
- Built the production agentic RAG pipeline for BEKO: document parsing, multilingual-e5 embeddings, query transformation (rewriting, HyDE, multi-query, sub-question decomposition), Qdrant vector retrieval and BGE cross-encoder reranking.
- Helped the system reach 98% tool-calling accuracy and 97% agentic RAG retrieval accuracy on a 300+ scenario evaluation set, validated via LLM-as-a-Judge and human review.
- Ran orchestration benchmarks for Bürotime's multi-agent platform comparing single-agent sequential, multi-agent sequential and concurrent patterns; the team adopted concurrent orchestration, reaching 97% correct agent selection, 98% correct tool calling and ~12 s responses on complex multi-step requests that could be parallelized.
- Contributed full-stack upstream PRs (Python, Vue) to SAGE, GT-ARC's open-source opaca-llm-ui, adding self-hosted model support, backend session management and UI improvements while building a Slack-to-SAGE integration.

AI/ML Engineer Intern, GT-ARC Gemeinnützige GmbH

Jul 2024 – Sep 2024

Berlin, Germany

- Engineered a text-to-video generation pipeline (open-source TTS + SadTalker, 512×512) and cut generation latency by 53% (5.3 s → 2.5 s) through pipeline-level optimizations such as parallelized stages and caching, while reaching a 7.29 SyncNet confidence score.
- Integrated GPT-3.5 into an interactive educational web UI, building an autonomous AI Tutor with adaptive Q&A that improved response coherence by 35%, as rated through human evaluation.

EDUCATION

Sabancı University, B.Sc. Computer Science and Engineering

2020 – 2025

Istanbul, Türkiye · Graduated June 2025 · Honor certificate (Fall 2021)

PROJECTS

Agentic Algorithmic Trading System on a Live Brokerage | GPT-4o, Python, C#/.NET, Kafka, gRPC,

Sep 2024 – Jun 2025

FinBERT, Docker, GitHub Actions

- Led development of an agentic BIST30 trading system on a live brokerage API (AlgoLab): ML signal models + a custom GPT-4o agent for data gathering and order decisions, C# execution over Kafka and gRPC; traded live with real capital after simulation.
- 65.41% simulated return, 52.8% directional accuracy, Sharpe 1.4, max drawdown -12%; Docker Compose, GitHub Actions (under 5 min to production); layered risk controls (limits, stop-loss, kill switch, HITL).

Cloud-Native Distributed To-Do Platform | FastAPI, AWS EC2, GCP, k3s, NGINX Ingress, Prometheus,

Feb 2025 – Jun 2025

Locust

- Deployed a FastAPI app on AWS EC2 and a self-managed k3s cluster on GCP VMs; load-tested to 2,000 concurrent users and tuned RPS-based pod autoscaling (2 → 10 pods, Prometheus Adapter), sustaining sub-second responses.
- Implemented TCP socket and RPC communication; Docker; GCP load balancer (L4) and NGINX Ingress (L7).

Wellmarkt E-Commerce Platform | Java, Spring Boot, Kafka, ReactJS, PostgreSQL, MongoDB, Docker

Sep 2024 – Jan 2025

- Built an event-driven Spring Boot e-commerce platform (4+ microservices, Kafka, 20+ endpoints, JWT) with ReactJS, PostgreSQL and MongoDB (RBAC, orders, stock) at 500 ms latency; Docker, Jira & Scrum.

CERTIFICATIONS

NVIDIA Deep Learning Institute: Building Transformer-Based NLP Applications, Building Conversational AI Applications, Fundamentals of Accelerated Computing with CUDA C/C++, Fundamentals of Deep Learning

IBM: Machine Learning with Python (with honors), Databases and SQL for Data Science with Python · **DeepLearning.AI:** Neural Networks and Deep Learning

SKILLS

Python, C/C++, C#, CUDA, SQL, LangGraph, RAG, Agentic AI, LLM Evaluation, FastAPI, Kafka, gRPC, Docker, Kubernetes, AWS, Qdrant, MongoDB, PostgreSQL

Languages: Turkish (native), English (professional working proficiency), German (A1–A2)